INCREMENTALITY STUDY · PORTFOLIO PROJECT 2

Did the Ads Actually Work?

What retargeting caused in a randomised test, and three decisions it informs: trust attribution, move budget, test next

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Independent analysis of public data: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) and Iowa Liquor Sales. Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

Executive summary

A retailer pays for retargeting: ads shown to people who have already visited its website. Its dashboards credit those ads with a large share of sales. The question is simple: **how many of those sales would have happened anyway?**

I answered it with a public randomised ad test from Criteo on 13,979,592 users: 85% were allowed to see the ads and 15% had them blocked, so the difference between the groups estimates what the ads caused. Criteo sub-sampled the file, so every level below describes this benchmark, not Criteo's live campaigns; the methods carry over.

Decision	What the evidence says	Recommendation
Should we trust attribution?	An estimated **54 of every 100** view-through-credited conversions were caused by the ads (95% interval 50 to 58). Across 7 adjustment methods: 51 to 58	Not on its own: attributed ROAS overstates incremental ROAS by about 1.9x
Should we move budget?	The top 20% of users hold 91% of the estimated effect; half the spend keeps 84% of caused conversions. Scenario (ROAS 8, margin 30%): the weaker half makes -£0.58 per £1	Probably, but test first: the split comes from a model
Which experiment next?	A 50/50 holdout inside the weaker half predicts a +55% lift against a +131% break-even	Run it (about 0.4 million users), then keep an always-on holdout

What else I found

- **The dashboard view is wildly misleading.** Users who saw an ad converted 41x more often than users who didn't. But they were already 19x likelier to buy; the estimated causal effect of seeing an ad is 2.2x.
- **The test needed a correction.** A small leak from merged advertiser tests inflated the raw lift (+59% for conversions, against +50% to +56% across the adjusted methods).
- **In the scenario, the dashboard overstates profit nearly fivefold:** each £1 of ads looks like +£1.40 but makes an incremental +£0.29. With a 30% margin, a campaign needs a dashboard ROAS above 6.2 to break even.
- **Geo tests are blunt (technical appendix).** On real Iowa sales with 2,700 simulated campaigns, even 20 counties for 12 weeks only reliably detected a lift of about 9%, and a CausalImpact-style model raised false alarms 10% to 18% of the time instead of 5%.

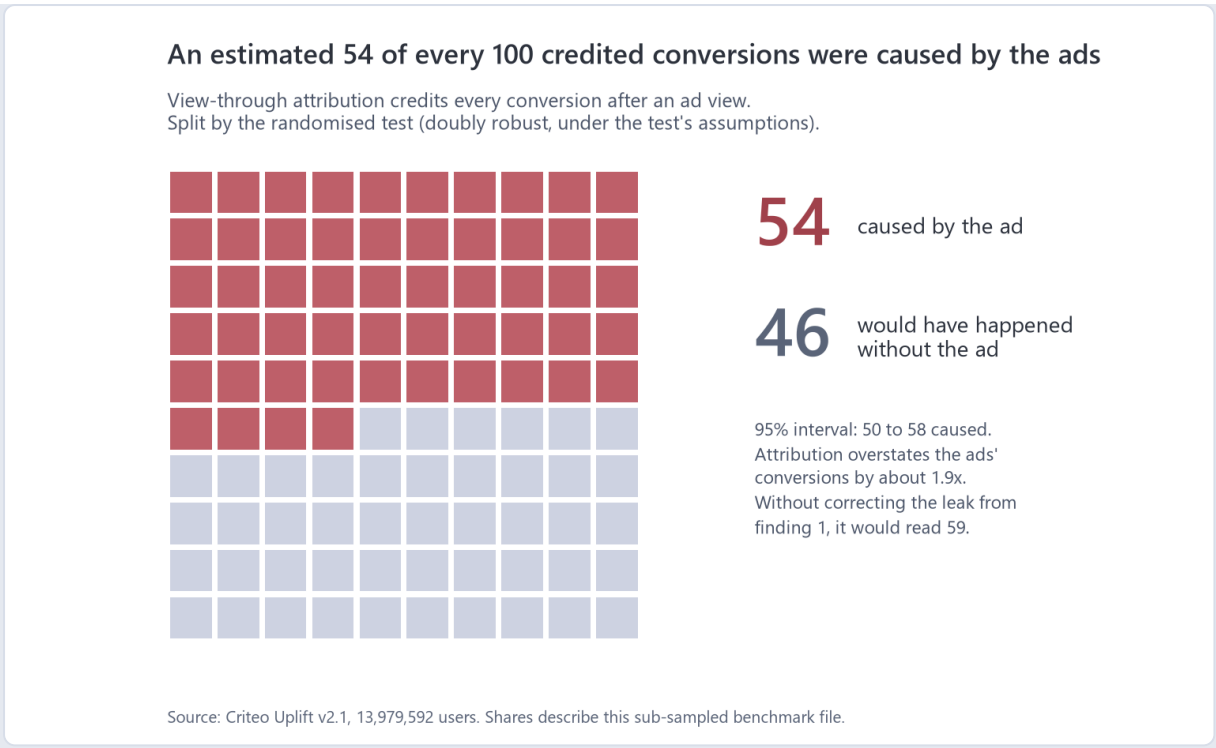

An estimated 54 of every 100 credited conversions were caused by the ads
View-through attribution credits every conversion after an ad view.
Split by the randomised test (doubly robust, under the test's assumptions).
54
caused by the ad
46
would have happened without the ad
95% interval: 50 to 58 caused.
Attribution overstates the ads' conversions by about 1.9x.
Without correcting the leak from finding 1, it would read 59.
Source: Criteo Uplift v2.1, 13,979,592 users. Shares describe this sub-sampled benchmark file.

Figure 1

1. The question and the data

Why attribution can't answer it

Attribution hands credit for a sale to the ads a person saw before buying. It cannot say whether the person would have bought anyway, and retargeting aims precisely at people who are already likely to buy. My previous project, a review of Google's Merchandise Store, showed attribution models disagree with each other and that its marketing mix model could not be calibrated without experiments. This project supplies the experiment.

The Criteo test

Criteo, an ad-tech company, ran randomised incrementality tests and released 13,979,592 users from them. For each user: which group they were in (85% treatment, 15% control), whether they actually saw an ad, whether they visited the advertiser's site and whether they converted, plus 12 anonymised features recorded before any ad was shown.

Only 3.6% of the treatment group actually saw an ad: being allowed to see ads doesn't mean you will. No control user saw one. That is what makes the effect of seeing an ad measurable.

Limits stated up front. Criteo sub-sampled the file so that its real incrementality can't be recovered, so the absolute numbers describe this benchmark file. There are no prices or costs, so money figures are scenarios with stated assumptions. The features are anonymised, so persuadable users can be described by behaviour but not by who they are. A second dataset, Iowa liquor sales, is used only in Appendix A to benchmark geo-test methods.

2. Is the test sound? (Finding 1)

Before trusting any lift number I checked the randomisation three ways.

- **The split** came out at exactly 85.00% treated, as designed.
- **Balance:** on every feature, treated and control users differ by less than 0.049 standard deviations, well inside the usual 0.1 line.
- **A classifier test:** if assignment were random, no model could tell the groups apart from the features. On 1 million users a model scored an AUC of 0.509, where 0.5 is a coin flip. A placebo with shuffled labels scored 0.499, so the small edge is genuine.

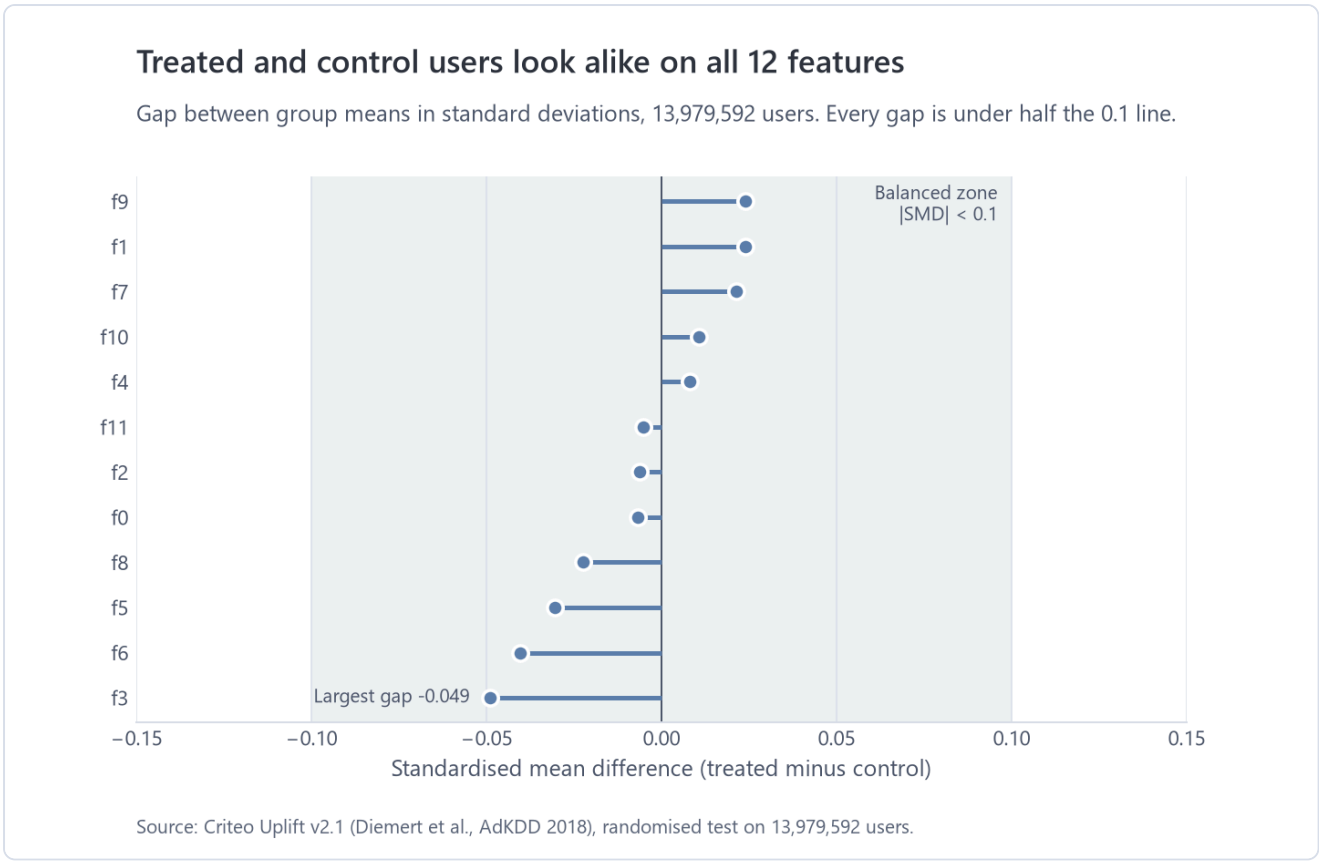


Figure 2

Criteo's own paper explains why: the file merges several advertisers' tests that had different treated shares. The first release leaked treatment information because of it; version 2 tried to fix it, and Criteo's check found no leak. At 14 million users, a small one remains.

The features were recorded at the moment users were assigned, before any ad, so adjusting for them cannot pick up any effect of the ads. The adjusted estimates are preferred, on one assumption that can't be tested: the leak runs through these recorded features. Section 4 shows how much the answer depends on the adjustment method.

3. Dashboards vs the test (Findings 2 to 4)

The dashboard view

A dashboard compares users who saw an ad with users who didn't: 5.38% against 0.13%, which looks like a 41x boost.

The estimated lift for the whole treatment group

Comparing the randomly assigned groups as a whole, the treatment group converted **52% more** than control and visited **20% more** (doubly robust, preferred). The raw comparison says +59% and +27%; the small leak from Finding 1 inflated it.

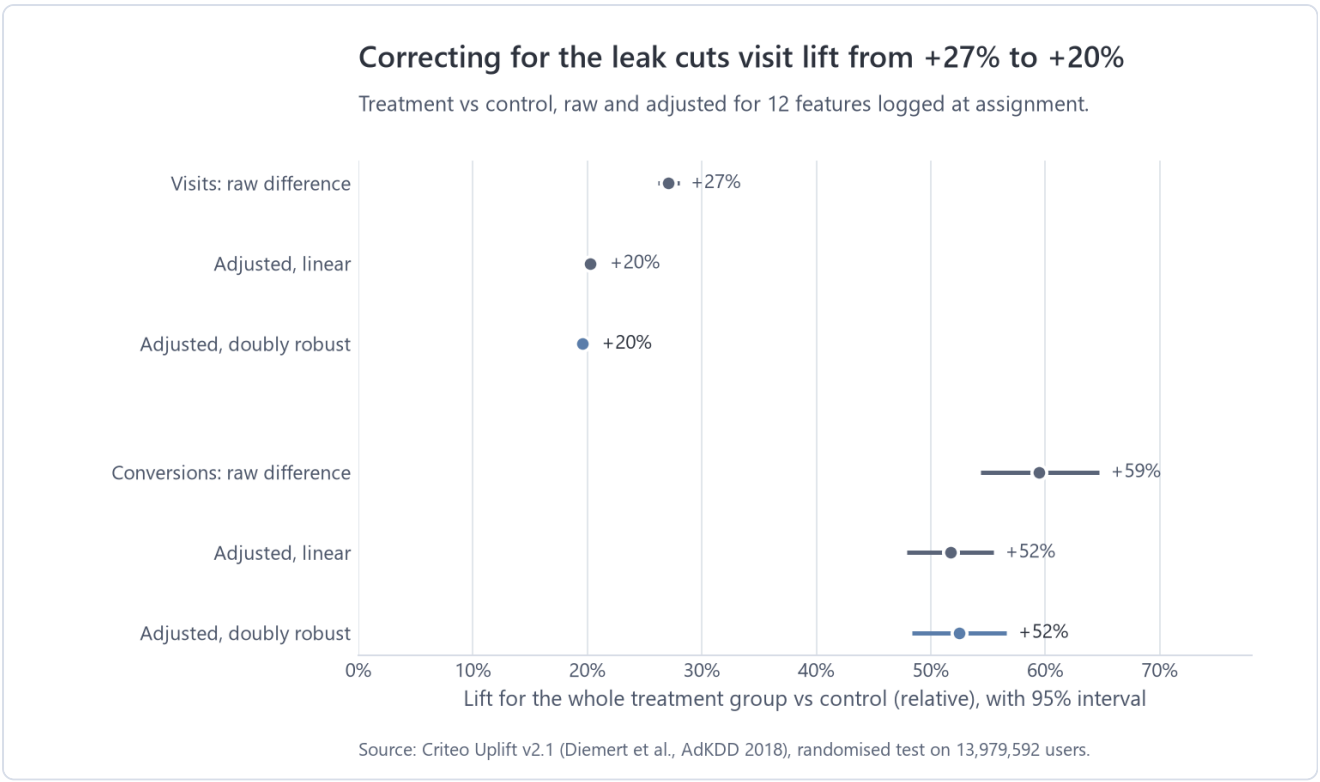


Figure 3

The effect of actually seeing an ad

Because nobody in control could see an ad, the random assignment works as an instrument: the whole-group lift divided by the share who saw an ad gives the effect on those who did (two-stage least squares gives the same number). Users who saw an ad converted at 5.38%; without it they would have converted at about 2.49%, already 19x the rate of everyone else. The estimated causal effect of the ad is **+2.89 points, or 2.2x**, not 41x. This needs the exclusion restriction: being in the treatment group changes behaviour only through seeing an ad.

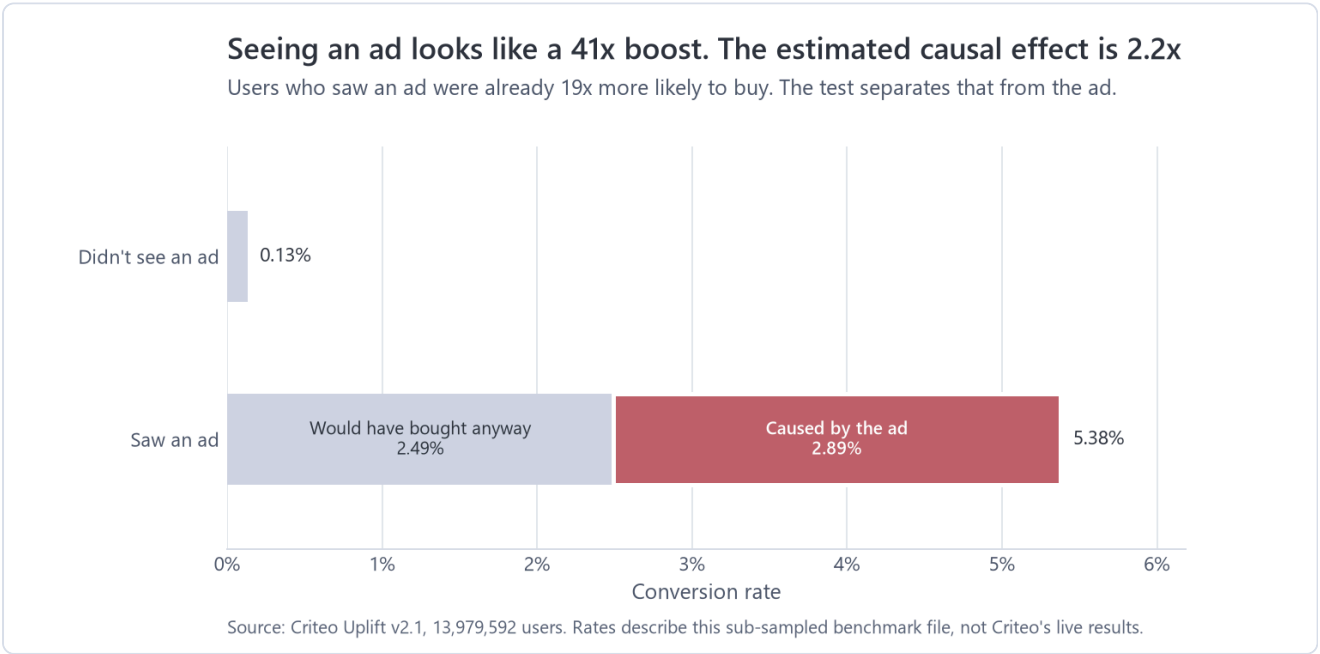


Figure 4

4. Decision 1: should we trust attribution? (Finding 5)

View-through attribution credits every conversion by a user who saw an ad. The test estimates that **54 of every 100 of those were caused by the ad** (95% interval 50 to 58); the other 46 would have happened anyway.

So an attribution report overstates retargeting's conversions, and its return on spend, by about **1.9x**: incremental ROAS = attributed ROAS x 0.54. Ads do work: seeing one roughly doubles a user's chance of buying. The problem is that attribution also hands them the sales of keen shoppers who were going to buy anyway. A click-based rule would credit fewer conversions, so its overstatement would be smaller.

How robust is the 54?

Without the Finding 1 correction the figure would read 59. To see how much the correction depends on modelling choices, I re-estimated the lift 7 ways: linear adjustment, and with two model families (gradient boosting and logistic regression) doubly robust, weighting only and outcome model only.

- Conversion lift across the adjusted specifications: **+50% to +56%**; visits +19% to +20%. Every adjusted estimate sits below the raw one.
- Caused per 100 credited conversions: **51 to 58**.
- Weighting by the fitted chance of treatment shrinks the largest feature gap from 0.049 to 0.002 standard deviations, so both propensity models capture the leak the features carry.

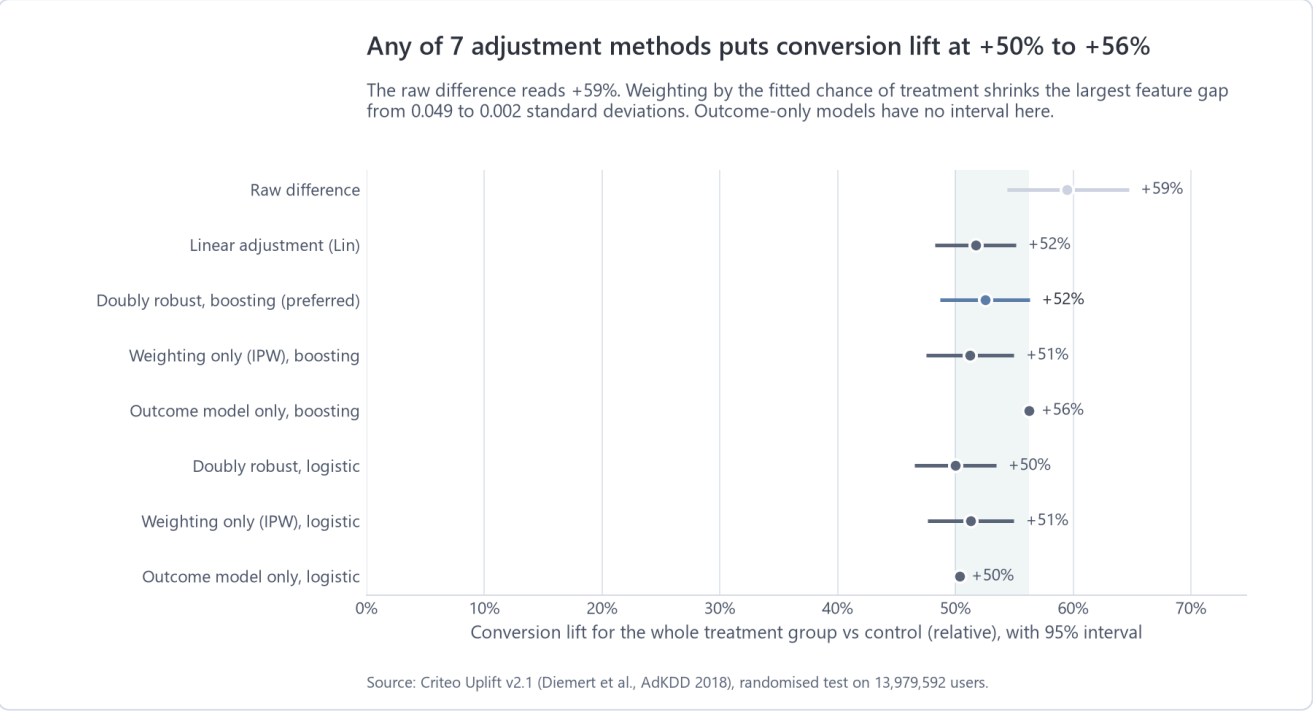


Figure 5

Recommendation: report incremental ROAS next to attributed ROAS in every campaign review, using a caused share measured by the business's own holdout.

5. Decision 2: should we move budget? (Findings 6 and 7)

Who is persuaded

Some users buy whether or not they see an ad; only persuadable users are worth paying to reach. I trained uplift models (a T-learner and an X-learner) on 80% of users and graded them on the other 2.8 million against a simple "likely buyers" model, the kind retargeting usually uses.

- **The effect is concentrated:** the top 20% of users hold about **91%** of it.
- **The simple model is as good as the advanced ones** for conversions. Both pick largely the same people, because only users who are shown ads can be persuaded and ads go to keen shoppers. The uplift model wins only for visits on a tight budget (top 10% of users: 80% vs 68% of the effect).
- **Measured by spend**, the yardstick that matters for money: half of today's ad spend, aimed at likely buyers, keeps **84%** of the conversions the ads cause (78% to 90%).

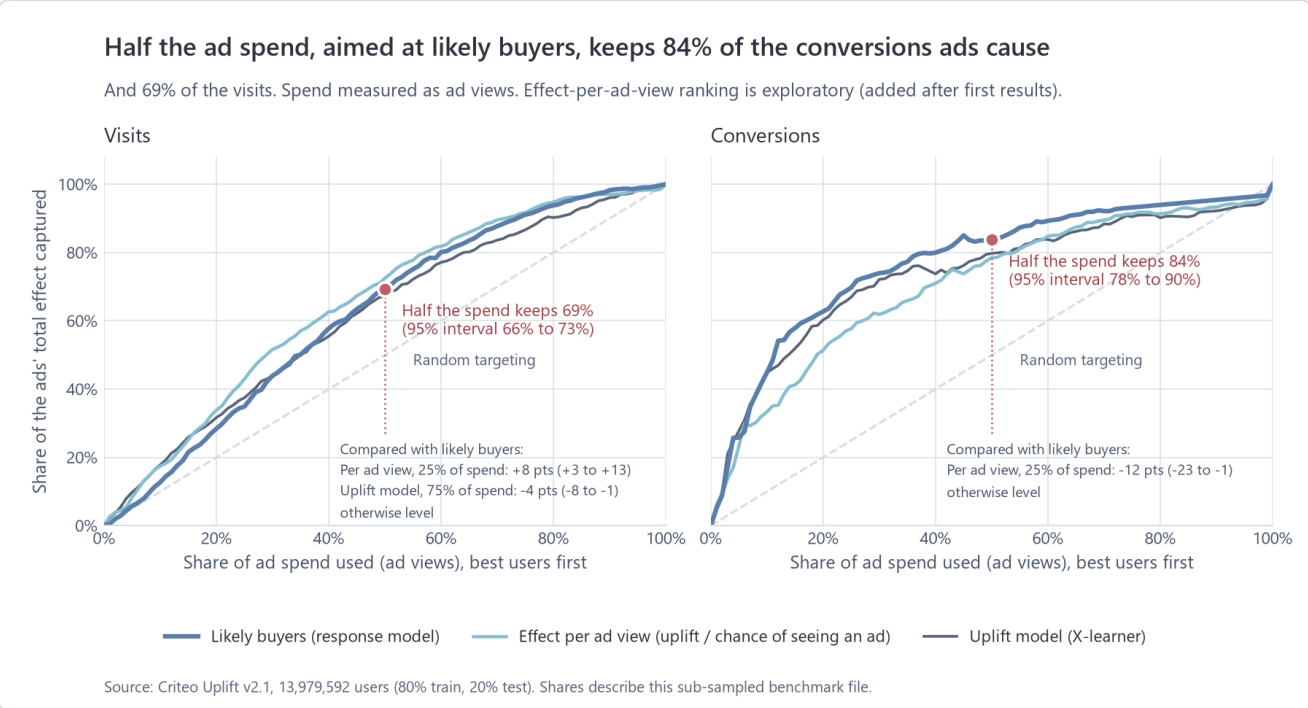


Figure 6

Is it worth the money? A scenario

The data has no prices or costs, so this is a scenario, not observed profit: a dashboard ROAS of 8 and a 30% margin, both easy to swap. The caused share and the spend split come from the test.

Profit per £1 of ad spend	Estimate (95% interval)
What the dashboard implies	+£1.40
Incremental, all spend	+£0.29 (+£0.19 to +£0.38)
Incremental, best half of spend	+£1.16 (+£0.94 to +£1.39)
Incremental, other half of spend	-£0.58 (-£0.74 to -£0.42)

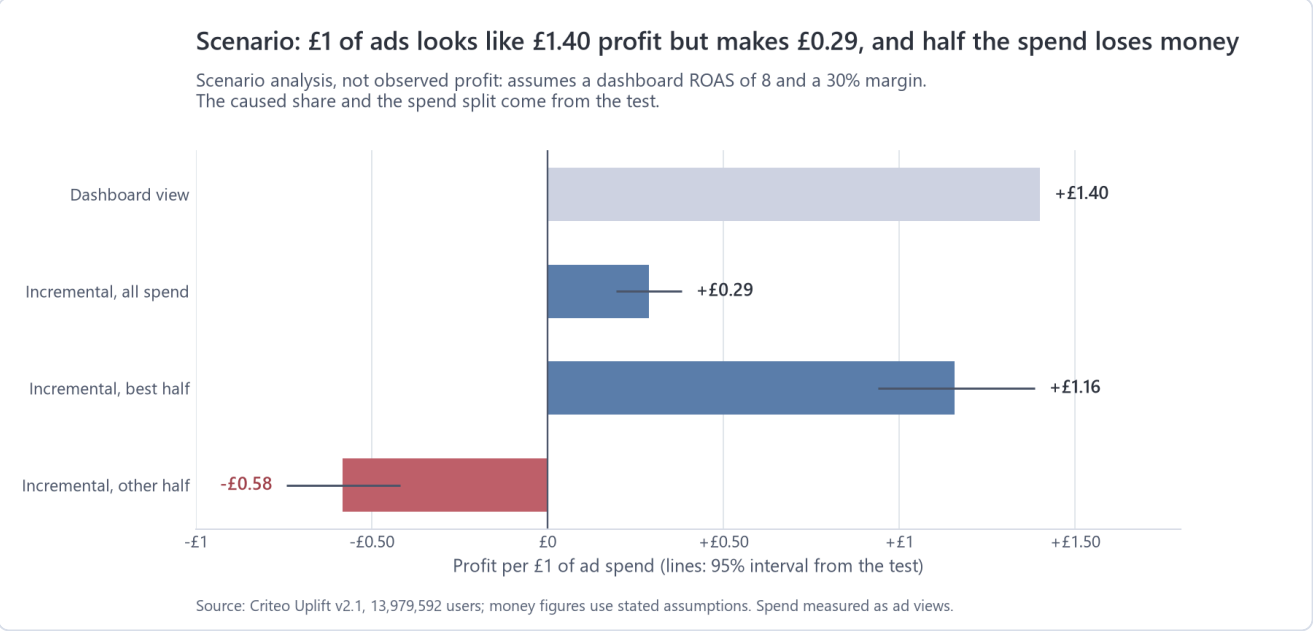


Figure 7

A rule that needs no ROAS assumption: **with a 30% margin, a campaign needs a dashboard ROAS above 6.2 to break even, and its weaker half needs 19.2.** Across dashboard ROAS of 4 to 15 and margins of 20% to 50%, the weaker half loses money in 21 of 24 combinations.

Why this is a hypothesis, not a saving

The split of spend comes from scoring users in a past test. Moving budget in production changes what that analysis held fixed: **auction prices** (fewer bids changes win rates and prices in the best half), **frequency** (the same budget on fewer users means more ads per user, with diminishing returns) and **audience composition** (new users, seasonality and model drift). Spend is also proxied by ad views. So the recommendation is to test the cut.

6. Decision 3: which experiment next? (Finding 9)

First: test the budget cut

	Proposal
Segment	Users ranked lowest by the likely-buyer model, up to half of today's ad views: about 89% of users
Design	Split the segment 50/50 into "keep ads" and "no ads"; leave the best half unchanged
Primary metric	Relative conversion lift in the segment; visits as the early read
Prediction	+55% (+33% to +76%)
Break-even	The segment pays for itself only at a lift of about +131% (scenario: ROAS 8, margin 30%)
Decision rule	Cut if the one-sided 95% upper bound on lift is below break-even; otherwise keep and re-rank
Size	About 0.4 million users gives an 80% chance of reaching "cut" if the prediction is right
Guardrails	Total conversions, cost per ad view and ads per user in the best half

Then: an always-on holdout, sized properly

I built a sizing calculator from the test's own rates and checked it against the precision the Criteo test achieved (within 0.2 points). Conversions are rare, so a 10% holdout needs about **4.9 million users** to see a +10% conversion lift (1.8 million with a 50/50 split). Visits give an early read with about 0.9 million.

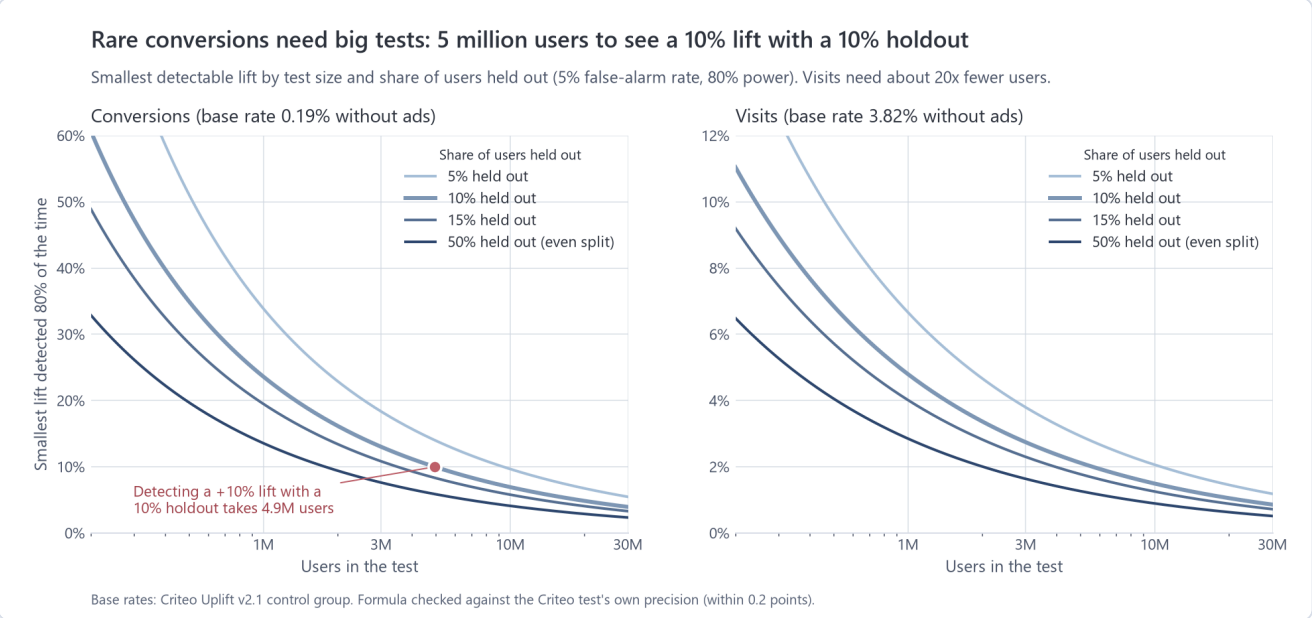


Figure 8

The rest of the plan

1. **Report incremental ROAS** (dashboard ROAS x caused share) next to attributed ROAS.
2. **Start with the simple likely-buyer model**; revisit uplift modelling with more data.
3. **Health-check every experiment** (split, balance, classifier with placebo), adjust for pre-treatment features and report how much the adjustment depends on the method.
4. **Use geo tests only where holdouts are impossible**, with difference-in-differences or synthetic control, sized by simulation and placebo-checked (Appendix A).
5. **Use a marketing mix model for cross-channel budgets, anchored by these experiments.**

Question	Best method	Watch out for
Is this digital campaign working?	User-level holdout	Holdout too small for rare conversions
Who should we target?	Likely-buyer or uplift model, checked by holdout	Grade by spend, not head count
Should we cut part of the budget?	Holdout on that part	Price and frequency shifts in the rest
Is a channel with no user-level control working?	Geo test (DiD or synthetic control)	Detects only large effects on noisy data
How should we split budget across channels?	MMM calibrated with experiments	Correlation unless anchored by tests

Before trusting any ad number, ask: is it attributed or incremental? Was the comparison randomised and health-checked? Could a test this size see an effect this big? What is the 95% interval? Has the method been placebo-tested?

7. Limits

- **Benchmark data.** Absolute numbers describe Criteo's sub-sampled file; every business must measure its own caused share. The methods transfer.
- **One untestable assumption behind the correction.** Adjusting removes the leak only if it runs through the 12 recorded features; every adjusted method agrees, but none can rule out a leak through something unrecorded.
- **Money is a scenario.** ROAS and margin are assumed; the break-even rule and sensitivity tables show how the conclusions move.
- **The budget split is model-based.** Production effects can change it; hence the proposed test.
- **Spend is proxied by ad views.** Real cost also depends on how many ads each user sees and their price.

Appendix A: geo tests, a methods benchmark (Finding 8)

TV, outdoor and some ad platforms can't hold out individual users. The fallback is a geo test: run the campaign in some regions, not others. **No real campaign is measured here.** I pulled weekly sales for 96 Iowa counties from BigQuery (2016 to 2026), added campaigns of known size myself and graded three methods on 2,700 simulated campaigns.

What the simulated campaign assumes: a uniform multiplicative lift in every test county and campaign week, test counties chosen at random, no spillover into control counties, no ramp-up or carryover, and 104 weeks of history. Real campaigns break each of these, so the results are a best case for this data.

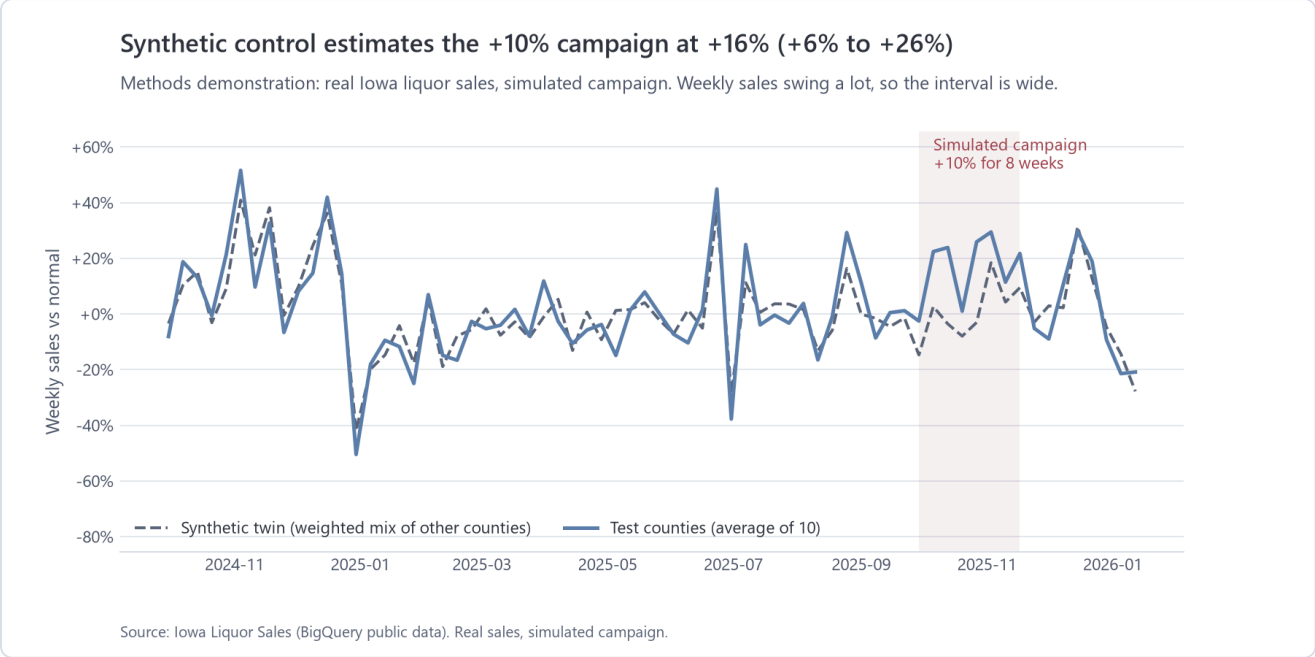


Figure 9

- **Two simple methods stay honest.** Difference-in-differences and synthetic control raised false alarms 3% to 8% and 4% to 7% of the time, close to the promised 5%.
- **The CausalImpact-style model cries wolf** 10% to 18% of the time, 2 to 3.5 times too often, and was the least accurate. (It is my own statsmodels build of the idea, not Google's package.)

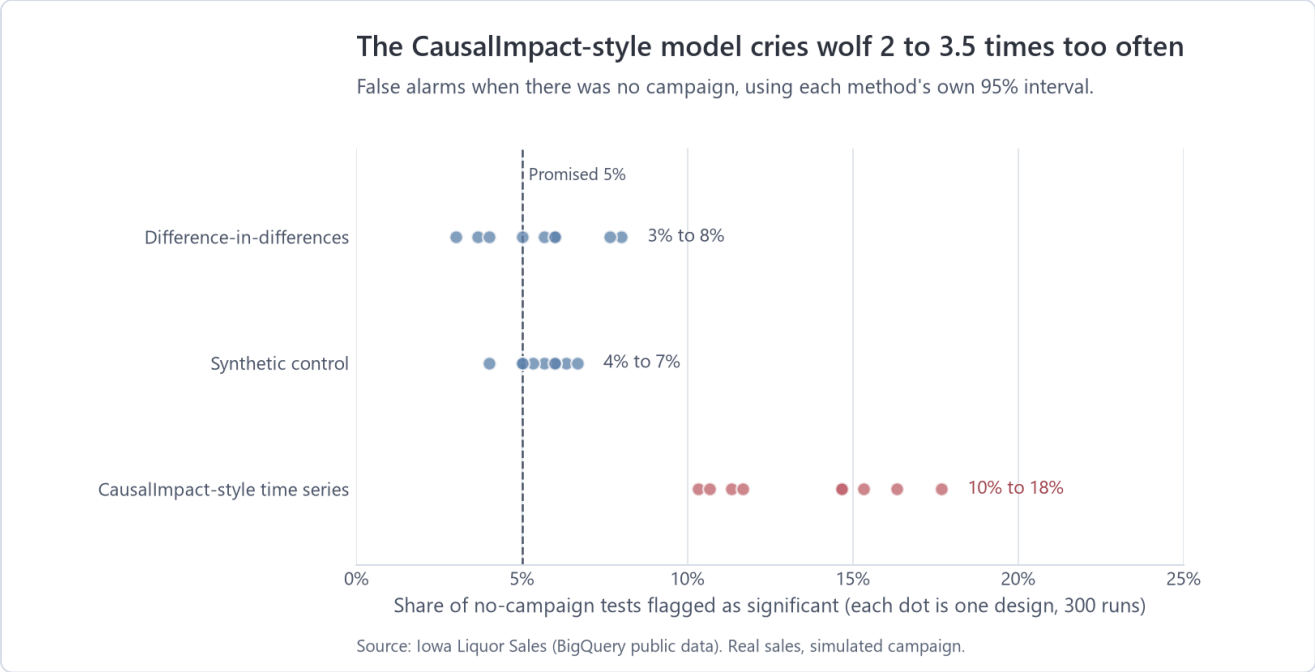


Figure 10

- **Only big effects are detectable.** Weekly county sales swing by about 50% from week to week. Even 20 counties for 12 weeks only reliably detect a lift of about 9%; 5 counties for 4 weeks need over 20%.

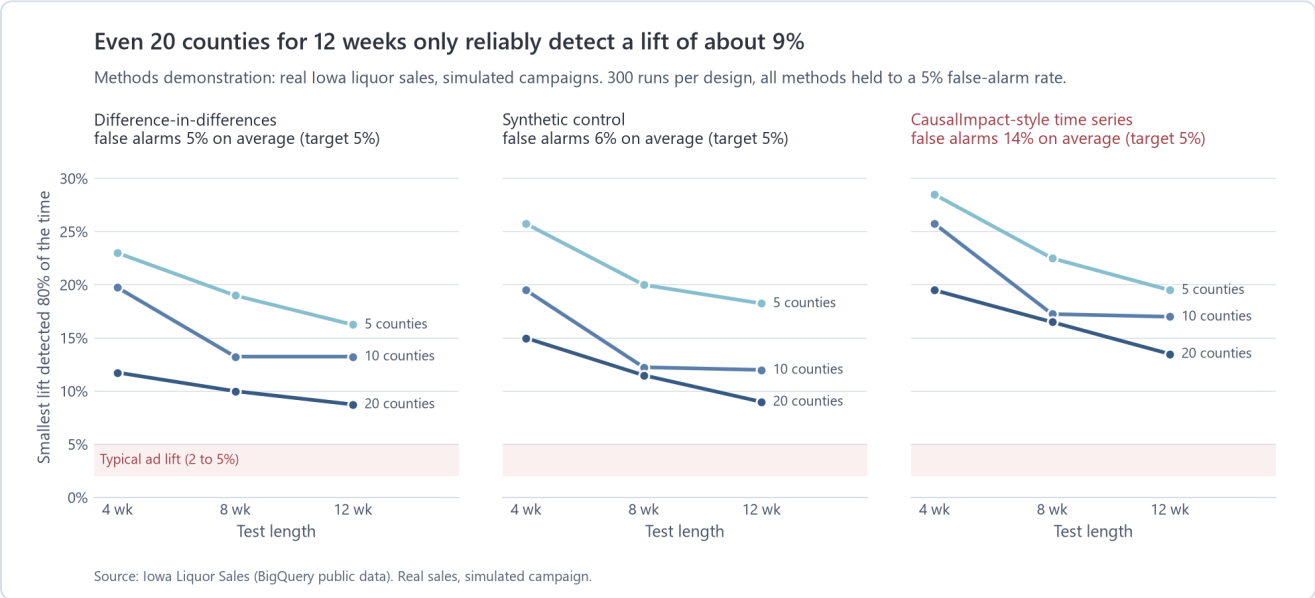


Figure 11

Appendix B: methods in brief

Step	Method	Code
Experiment health	Chi-square sample ratio test; standardised mean differences; gradient-boosted classifier two-sample test with placebo	`experiment.py`
Whole-group lift	Difference in proportions with 95% intervals; Lin (2013) regression adjustment; doubly robust AIPW with 2-fold cross-fitting	`lift.py`
Sensitivity	7 adjusted specifications: Lin, and AIPW, IPW and outcome-only with boosting and logistic models; weighted balance	`sensitivity_analysis.py`
Effect of seeing an ad	Wald / two-stage least squares (linearmodels), robust errors	`lift.py`
Uplift	T- and X-learner (gradient boosting), graded on held-out users with doubly robust pseudo-outcomes, paired intervals, tie-aware	`uplift.py`
Money	Incremental ROAS and profit per £, uncertainty by simulation	`money.py`
Budget test	Segment lift and break-even from held-out scores; one-sided sizing on the log-ratio scale	`budget_test_plan.py`
Geo tests	Collapsed difference-in-differences (Bertrand, Duflo and Mullainathan 2004); synthetic control with placebo inference; structural time series (statsmodels)	`geo.py`
Sizing	Minimum detectable relative lift on the log-ratio scale	`planning.py`

Appendix C: reproduce

Everything is rebuilt by `python python/scripts/run_all.py`; every number in this report is printed by `python python/scripts/headline_facts.py`. Code, notebooks and write-ups per finding are in the repository.

Appendix D: glossary

- **Incrementality:** sales caused by the ads, beyond what would have happened anyway.
- **Attribution:** rules that hand out credit for a sale to the ads a person saw before buying.
- **Intent-to-treat:** treatment group minus control group, as randomly assigned.
- **Instrumental variable:** something random that changes who gets treated, used to isolate the treatment's effect.
- **Doubly robust:** an estimate that stays right if either its model of who gets treated or its outcome model is right.
- **Uplift model:** a model that predicts how much an ad changes each person's chance of buying.
- **Geo test:** an experiment in which regions, not people, are the test and control groups.
- **Minimum detectable lift:** the smallest real effect a test catches 80% of the time.